Status	Finished
Started	Monday, 5 October 2026, 4:55 PM
Completed	Monday, 5 October 2026, 5:06 PM
Duration	10 mins 33 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n ✓
- b. log n
- c. n-1
- d. n/2

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is unsorted
- b. When the list is very small
- c. When the list is sorted ✓
- d. When the list contains duplicate elements

Question **3**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search can find multiple instances of the target element
- b. Binary search has a lower time complexity on large, sorted lists ✓
- c. Binary search does not require dividing the list
- d. Binary search works on unsorted lists

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. The list must be sorted

b. There must be a mechanism to delete and/or insert elements in the list ✓

c. There should be direct access to the middle element in any sublist

d. Number values should only be present

Number values should only be present

Question **5**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 99 ✓

b. 89 and 94

c. 90 and 100

d. 94 and 99

Question **6**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O👎 ✓

b. O(1)

c. O(n log n)

d. O(log n)

Question **7**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. At the beginning of the list
b. In the right sublist
c. In the left sublist ✓
d. In the entire list

Question **8**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element
b. By comparing each element sequentially
c. By dividing the list length by two ✓
d. By using a hash function

Question **9**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the search algorithm finishes
b. When the list is sorted
c. When the element is found in the list ✓
d. When the list contains duplicate elements

Question **10**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised
b. The first element is returned
c. The last element is returned
d. The search is considered unsuccessful ✓

Question **11**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Linear search
b. Binary search ✓
c. Hash search
d. Sequential search

Question **12**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99
b. 90 and 99 ✓
c. 90 and 94
d. 89 and 94

Question **13**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is somewhere in the middle of the array ✓
b. When the item is not the array at all
c. When the item is the last element in the array
d. Item is the last element in the array or item is not there at all

Question **14**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Bubble ✓

b. Complexity

c. Insertion

d. Selection

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Sorting the list before searching

b. Checking each element sequentially ✓

c. Dividing the list in half repeatedly

d. Skipping every second element